

Radiation—A Concern, iDose⁴—The Solution

By their extensive research in the field of computed tomography (CT), Philips Electronics India has come up with its latest technology, iDose⁴. With increasing number of patients undergoing CT scans every year and the growing concern of radiation exposure to such patients, iDose⁴ aims at reducing radiation dose in its new and existing CT products

ABHISHEK MUTHA

Diagnostic radiology tests utilising various imaging equipment like CT/X-ray are essential to arrive at an accurate diagnosis. iDose⁴ is an iterative-based reconstruction technology introduced by Philips in its CT scanners through which the scanner is enabled to reduce radiation dose up to 80 per cent without compromising on the quality of images. iDose⁴ is also capable of improving the image quality at low radiation dose. "This technology is amongst the most successful low-dose imaging technologies in the CT industry, which got widely appreciated by the entire radiology community," says Raveendran Gandhi, senior director-Radiology, Philips Electronics, who shared more details on this technology.

Radiation—A concern and the reason behind iDose⁴

Talking about the effects of radiation exposure, Raveendran Gandhi said, "Exposure to ionising radiation leads to several biological side effects. Multiple clinical studies on radiation effects on foetus have consistently found an association between exposure to medical radiation during pregnancy and risk of childhood cancer in the offspring."

According to Raveendran Gandhi, several clinical studies basing radiation effects on pediatric CT risks suggest that pediatric CT will result in significantly increased lifetime radiation risk over adult CT, both because of the in-

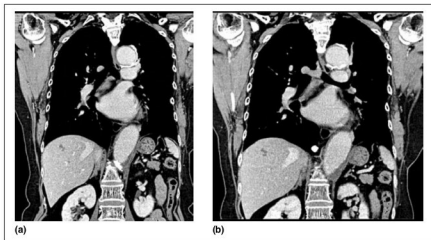


Fig. 1: (a) Full dose images (FBP - filtered back projection), (b) iDose⁴ images (80% less dose)

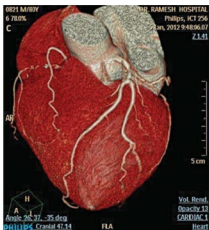


Fig. 2: iDose⁴ images of the heart showing calcification and narrowing of coronary arteries

creased dose per milliamper-second, and the increased lifetime risk per unit dose. In the UK, it has been estimated that approximately four per cent of diagnostic radiology procedures are CT examinations, but, their contribution to the collective radiation dose is

approximately 40 per cent.

This emphasises the need for low-dose CT imaging technology.

iDose⁴—A different approach to image reconstruction

"For decades, filtered back projection (FBP) has been the industry standard for CT image reconstruction. While it is a very fast and fairly robust method, FBP is a sub-optimal algorithm choice for poorly sampled data or for cases where noise overwhelms the image signal. Such situations may occur in low-dose or tube-power-limited acquisitions, for instance in the scans of morbidly obese individuals," says Raveendran Gandhi. Talking about the noise, he further adds, "Noise in CT projection data is dominated by photon count statistics. As the dose is lowered, the variance in the photon count statistics increases disproportionately.